

- 2 (a) The enzyme pectinase is used in the production of apple juice.

In an experiment, a student investigated the production of apple juice using five different concentrations of pectinase solution.

For each concentration, the student:

- put 100g of crushed apple into a beaker
- added 10 cm³ of the pectinase solution to the crushed apple
- left the crushed apple and enzyme mixture for 30 minutes at 40 °C
- filtered the apple juice from the mixture
- measured the volume of apple juice produced.

- (i) The temperature was maintained at 40 °C.

Describe how the student could maintain the crushed apple and enzyme mixture at a constant temperature.

.....

 [1]

- (ii) State **two other** variables that were kept constant in this investigation.

1

 2
 [2]

- (b) The results of this investigation are shown in Table 2.1.

Table 2.1

percentage concentration of the pectinase solution	volume of apple juice produced / cm ³
0.1	4
0.2	6
0.4	12
0.8	22
1.6	22

